

Foundational Genetic and Protein Language Models: A Taxonomy of Training Data, Architecture, Scaling Laws, and the Data Bottleneck

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bioRxiv preprint * September 2026 * Companion survey to LewyGym (github.com/Tyronita/LewyGym)

We survey 25 foundational biological language models spanning protein sequence (ESM family, ProtTrans, ProGen/ProGen2, RITA, AMPLIFY, Tranception, EVE, SaProt, AlphaMissense, ESM-IF), DNA/genomic (DNABERT/2, Nucleotide Transformer v1/v2, HyenaDNA, Caduceus, Evo, Evo 2), and multimodal (ESM-3) modalities, cataloguing their training datasets, architectures, parameter counts, perplexity at scale, and downstream benchmark performance. We find that: (1) protein MLM scaling exponents ($N_{\text{opt}} \sim C^{0.77}$) differ substantially from text LLMs ($\sim C^{0.50}$), meaning existing large protein models are systematically undertrained; (2) zero-shot fitness prediction on ProteinGym peaks at ~650M-4B parameters and then declines, with a parameter-free evolutionary statistics method (RSALOR, $\rho=0.582$) outperforming all neural models including 40B-parameter Evo 2 ($\rho=0.360$); (3) genomic DNA models have solved the long-range context problem via subquadratic architectures (StripedHyena, Mamba), with Evo 2 achieving 1M single-nucleotide context; and (4) the primary bottleneck is not compute or architecture but data diversity -- the 1.17B truly novel MGnify dark proteome sequences remain largely untapped, the UniRef diversity ratio declines yearly, and no RNA foundation model exists. We frame these findings in the context of disease-specific protein variant prediction, motivated by the LewyGym benchmark showing that gain-of-function proteins systematically invert conservation-based scoring signals.

Keywords: protein language model, genomic language model, scaling laws, deep mutational scanning, variant effect prediction, data bottleneck

1. Introduction

Biological language models -- pretrained neural networks operating on sequences of amino acids or nucleotides -- have emerged as the dominant paradigm for zero-shot prediction of protein fitness, variant pathogenicity, and molecular function [1-4]. The field has bifurcated into two complementary tracks: protein language models (PLMs) operating on amino acid vocabularies of 20-441 tokens, and genomic DNA/RNA language models operating on nucleotide vocabularies of 4-4096 tokens. Both tracks have rapidly scaled from hundreds of millions to hundreds of billions of parameters between 2021 and 2026.

Despite this rapid growth, no comprehensive survey has examined the full landscape of training datasets, architectural choices, scaling law behaviour, and fundamental bottlenecks across both tracks simultaneously. Existing reviews focus on either protein [5] or genomic [6] models separately and do not systematically characterise the data-efficiency frontier or identify the point at which additional compute ceases to improve downstream task performance.

This survey is motivated by an empirical finding from our companion work, LewyGym [7]: ESM-2 650M achieves Spearman $\rho = -0.17$ on alpha-synuclein (SNCA) variant fitness prediction, compared to $+0.414$ on ProteinGym's loss-of-function benchmark. Understanding why this inversion occurs -- and what training signals would be required to correct it -- requires understanding the full landscape of available models, datasets, and their respective limitations.

2. Protein Language Models

2.1 ESM Family (Meta AI / EvolutionaryScale)

The ESM series represents the most systematically studied protein LM scaling trajectory. ESM-1 (2021) established that biological structure and function emerge from masked language modelling on 250M protein sequences without structural supervision [1]. ESM-1b introduced Pre-LN stabilisation at 650M parameters. ESM-1v redirected the architecture toward variant effect prediction by training on the more diverse UniRef90 (98M sequences) rather than UniRef50, achieving Spearman $\rho = 0.510$ across 41 DMS datasets with an ensemble of 5 models [2]. ESM-2 replaced absolute position embeddings with Rotary Position Embeddings (RoPE), removed dropout entirely, and systematically scaled from 8M to 15B parameters, establishing the first clean protein-specific scaling law ($r = -0.99$ between perplexity and CASP14 TM-score across all six scales) [1]. ESM-3 (2024) extended the framework to joint multimodal generation over sequence, structure, and function tokens, training on 3.15 billion proteins and 771 billion tokens [3]. ESM Cambrian (2024) demonstrated that modern architecture (RoPE + SwiGLU + no biases) plus metagenomic data allows the 300M-parameter ESM C to match the 650M ESM-2, training on 6.2 trillion tokens including 2 billion JGI metagenomic clusters.

2.2 ProtTrans Family (Rostlab / TU Munich)

The ProtTrans suite [8] systematically evaluated four NLP architectures (BERT, ALBERT, XLNet, T5) on protein sequences, providing the first large-scale evidence that (a) bidirectional MLM outperforms autoregressive permutation LM (ProtXLNet Q3=73 vs ProtBERT Q3=75-76), and (b) metagenomic data (BFD, 2.1B sequences, 393B tokens) consistently improves over curated databases (UniRef100, 216M sequences) across all downstream tasks. ProtT5-XL (3B) trained on BFD then fine-tuned on UniRef50 achieves Q3 secondary structure = 81, the best single-sequence result in the suite.

2.3 Generative Protein Models (ProGen, ProGen2, RITA, ProtGPT2)

Generative protein models take a GPT-style decoder-only autoregressive approach. ProGen (Salesforce, 1.2B, 2020) introduced conditional generation via control tags. RITA (LightOn, 2022) provided the most systematic autoregressive scaling study: 85M to 1.2B parameters on UniRef-100 with RoPE, finding a scaling exponent of 0.74 (loss vs model size) -- steeper than natural language (~ 0.50) -- and perplexity 5.48 on UniRef-100 for RITA-XL [9]. ProGen2 (Salesforce, 2022) scaled to 6.4B parameters on UniRef90 + BFD30 using GPT-J with RoPE, achieving monotonic perplexity improvement (small 12.9 $\rightarrow$ xlarge 9.9) but non-monotonic zero-shot fitness -- ProGen2-base (764M) marginally outperforms xlarge (6.4B) on ProteinGym Spearman (0.505 vs 0.476) [10]. ProtGPT2 (738M, 2022) showed that GPT-2 style training on 44.9M UniRef50 sequences generates structurally plausible proteins but achieves poor perplexity (18.10 on UniRef-100 vs RITA-XL 5.48).

2.4 Efficiency Frontier (AMPLIFY, xTrimoPGLM)

AMPLIFY (Mila / Amgen, 2024) demonstrated that modern architectural components (RMSNorm, SwiGLU, RoPE, FlashAttention) combined with deduplicated training data (UR100P: 391M sequences) allow AMPLIFY-350M to outperform ESM-2 15B on ProteinGym benchmarks with 43x fewer parameters and 17x fewer FLOPs [11]. xTrimoPGLM (Tsinghua / BAAI, 2024) scaled a GLM encoder-decoder to 100B parameters on ~ 1 trillion tokens, achieving state-of-the-art across 18 benchmarks.

Model	Yr	Parameters	Training Data	Perplexity	ProtGym rho	Arch
ESM-1 / ESM-1b	2021	43M-670M	UniRef50 250M seqs	n/r	-	BERT

ESM-1v	2021	650M x5	UniRef90 98M seqs	7.29	0.510	BERT
MSA Transformer	2021	100M	UniRef50 MSAs	n/r	0.434	Axial
ESM-2	2022	8M-15B	UniRef50/90 ~1T tok	6.37@15B	0.414	BERT+RoPE
ProtBERT / BFD	2021	420M	UniRef100 / BFD 393B ..	n/r	-	BERT
ProtT5-XL	2021	3B / 11B	BFD 393B, UR50 finetune	n/r	0.51*	T5+Enc
ProtGPT2	2022	738M	UniRef50 44.9M seqs	18.10*	-	GPT-2
RITA-XL	2022	1.2B	UniRef-100 280M seqs	5.48	-	GPT+RoPE
ProGen2-xlarge	2022	6.4B	UR90 + BFD30 ~180M seqs	9.9	0.476	GPT-J+RoPE
Tranception-L	2022	700M	UniRef100 250M seqs	n/r	0.457*	GPT+conv
EVE	2021	~M / protein	Per-protein MSAs	n/r (ELBO)	0.441	VAE
AlphaMissense	2023	n/d (AF2-scale)	AF2 data + gnomAD pop..	n/a (clf)	0.474	AF2 fine..
SaProt-650M	2023	650M	40M AF2 structs (3Di ..	n/r	0.478	ESM-2+SA..
ESM-3 (1.4B open)	2024	1.4B-98B	3.15B seqs, 771B tokens	0.895 LDDT	0.466	Multi-mo..
ESM C (6B)	2024	300M-6B	6.2T tok incl JGI 2B	n/r	>ESM-2	BERT+RoPE
AMPLIFY-350M	2024	120M / 350M	UR100P 391M seqs	n/r	>ESM-2 15B	BERT+RMS..
xTrimoPGLM	2024	100B	UR90+ColdFold ~1T tok..	10.81	SOTA	GLM-100B

Table 1. Protein LMs surveyed. ProtGym rho = mean Spearman on ProteinGym substitution benchmark (zero-shot). n/r = not reported; n/d = not disclosed; n/a = not applicable; * = VESPA/ProtT5 proxy; Tranception* = with retrieval; EVE is per-protein, not a global model. Arch: BERT = bidirectional MLM encoder; GPT = autoregressive decoder; T5 = encoder-decoder; RoPE = Rotary Position Embeddings; SA = structure-aware vocabulary.

3. DNA and Genomic Language Models

DNA language models face a qualitatively different challenge from protein LMs: the genomic sequence space is less information-dense (4-letter alphabet vs 20), regulatory signals operate at long range (enhancers 100kb-1Mb from target genes), and the same sequence encodes multiple overlapping functional elements in both strands. Early models (DNABERT, 2021) applied BERT to k-mers from the single human reference genome, demonstrating chromatin accessibility (AUROC 0.85) and methylation prediction. The dominant recent trend is architectural: subquadratic alternatives to attention (Hyena, Mamba/SSM) enable single-nucleotide context of 1M+ base pairs, which transformers cannot achieve at any practical scale.

3.1 k-mer to Single-Nucleotide Tokenization

DNABERT (2021) used k-mer tokenization (k=3-6) with 512-token context (~3kb), limiting coverage to short regulatory elements. DNABERT-2 (2023) replaced fixed k-mers with BPE tokenization over 135 species (32.5B nucleotides), achieving 5x compression and enabling context extension; at 117M parameters it matches the 2.5B Nucleotide Transformer on 28-task GUE benchmark using 21x fewer parameters and 92x less GPU time. Nucleotide Transformer v1 (InstaDeep/EMBL-EBI, 2023) scaled to 2.5B parameters with 6-mer tokenization over 850 diverse genomes; NT-v2 (2023) replaced learned absolute positions with RoPE and SwiGLU, matching NT-v1-2500M performance at 5x smaller model size.

3.2 Subquadratic Architectures: Hyena and Mamba

HyenaDNA (Hazy Research, 2023) replaced self-attention with Hyena implicit convolutions ($O(L \log L)$), enabling 1M-bp single-nucleotide context at only 6.6M parameters. Trained on the single human reference genome, it outperforms models 1,500x larger on most GenomicBenchmarks tasks by exploiting long-range dependencies inaccessible to k-mer attention models [12]. Caduceus (Cornell, 2024) added bidirectionality and reverse-complement equivariance to Mamba/SSM blocks, improving HyenaDNA on double-stranded DNA tasks by encoding both strands symmetrically.

3.3 Genomic Foundation Models at Scale (Evo, Evo 2)

Evo (Arc Institute, 2024, 7B parameters) trained StripedHyena on 300B tokens of prokaryotic and phage genomes (OpenGenome), enabling zero-shot CRISPR guide RNA activity and gene essentiality prediction [13]. Evo 2 (2025, up to 40B parameters) extended to all three domains of life via OpenGenome2 (8.85T nucleotides, 15,032 eukaryotic + 113,379 prokaryotic genomes), achieving 1M single-nucleotide context and state-of-the-art noncoding variant effect prediction (ClinVar noncoding AUROC, ranked 1st). Perplexity 2.597 at 1M context (7B), vs Transformer baseline 3.09 at identical scale.

Model	Yr	Parameters	Tokenization	Training Data	Context	Architecture
DNABERT	2021	~86-89M	k-mer (k=3-6)	hg38 (3.2B nt)	512 tok ..	BERT, fixed pos
DNABERT-2	2023	117M	BPE (4096 vo..)	135 species 32.5B nt	Ext. (AL..)	MosaicBERT+ALiBi
NT v1-2500M-multi	2023	500M/2.5B	6-mer (4105 ..)	850 genomes 174B nt	6kb (100..)	BERT, learned pos
NT v2-500M	2023	50M-2.5B	6-mer	Human+multi 300B-1T tok	12kb (20..)	BERT+RoPE+SwiGLU
HyenaDNA	2023	0.4M-6.6M	Single-nucle..	hg38 only (~3B nt)	1M bp	Hyena (O(L logL))
Caduceus	2024	~7.7M	Single-nucle..	hg38 (~3B nt)	131,072 nt	BiMamba+RC equiv
Evo (Evo 1)	2024	7B	Single-nucle..	OpenGenome 300B tok	131,072 nt	StripedHyena
Evo 2	2025	1B-40B	Single-nucle..	OpenGenome2 8.85T nt	1,048,57..	StripedHyena 2

Table 2. RC equiv = reverse-complement equivariant; BiMamba = bidirectional Mamba/SSM.

4. Architecture Choices and Their Consequences

4.1 BERT vs GPT vs T5 for Proteins

The protein modelling literature has decisively favoured bidirectional MLM encoders (BERT-style) over autoregressive decoders (GPT-style) for representation learning and zero-shot fitness prediction. ProtXLNet (autoregressive permutation LM) underperforms ProtBERT (bidirectional MLM) across all tasks in the ProtTrans suite [8], and ProGen2-xlarge (6.4B GPT) underperforms AMPLIFY-350M (modern BERT) on ProteinGym despite being 18x larger. The T5 encoder-decoder hybrid performs well for per-residue prediction tasks (ProtT5 Q3 = 81) but is rarely used for generation tasks due to training complexity. For protein generation, GPT-style models are necessary and have scaled effectively (ProGen2, RITA).

4.2 Position Embeddings: Absolute vs RoPE vs ALiBi

Rotary Position Embeddings (RoPE) have become the default for all recent protein and genomic models. ESM-2 introduced RoPE to the protein domain; RITA, ProGen2, NT-v2, and Evo 2 all use RoPE. ALiBi (additive bias in attention scores) is used by DNABERT-2 to allow context-length extrapolation beyond training length. Learned absolute position embeddings (ESM-1b, DNABERT, NT-v1) hard-code maximum context length and impair extrapolation. The shift to RoPE enabled NT-v2 to double context from 6kb to 12kb relative to NT-v1 without additional parameters.

4.3 Activation Functions and Normalisation

AMPLIFY's analysis isolates the architecture improvement specifically: replacing GeLU with SwiGLU and LayerNorm with RMSNorm, combined with removing biases, yields AMPLIFY-350M outperforming ESM-2 15B at 43x fewer parameters. The gap is partly attributable to FLOPs efficiency (17x fewer) and partly to data quality (UR100P vs the 2021 UniRef50 release used by ESM-2). Disentangling architecture from data quality improvements remains difficult.

4.4 Subquadratic Architectures for Long Context

Transformers require $O(L^2)$ memory and compute for context length L , preventing practical use beyond $\sim 16k$ tokens. The Hyena operator (implicit long convolutions, $O(L \log L)$) enables HyenaDNA's 1M-bp single-nucleotide context at 6.6M parameters. Mamba/SSM-based models (Caduceus) provide linear-time sequence modelling with selective state transitions, adding reverse-complement equivariance for genomics. StripedHyena (Evo 1/2) interleaves Hyena convolutions with sparse MHA layers, running 3x faster than pure Transformers at 1M context while improving perplexity (2.597 vs Transformer baseline 3.09 at 7B scale).

4.5 Structure-Aware Vocabulary (SaProt)

SaProt (2023) achieved the most parameter-efficient improvement on ProteinGym ($\rho = 0.478$, ranked #1) by a single architectural change: replacing the standard 20-amino-acid vocabulary with 441 structure-aware (SA) tokens pairing each residue with its Foldseek 3Di structural state. No changes to the ESM-2 transformer architecture. This demonstrates that vocabulary design is an underexplored efficiency dimension: the same number of parameters encodes qualitatively richer information when tokens carry structural context.

5. Scaling Laws: Perplexity, Parameters, and Data Tokens

5.1 ESM-2 Protein Scaling Law

ESM-2 provides the cleanest protein-specific scaling law: validation perplexity decreases from 10.45 (8M parameters) to 6.37 (15B parameters) with Pearson $r = -0.99$ between $\log(\text{perplexity})$ and downstream CASP14 TM-score across all six scale points [1]. This near-perfect correlation confirms that perplexity is a reliable proxy for structural accuracy in protein MLMs -- a result not guaranteed a priori and not observed in natural language.

Model	Parameters	Perplexity	Contact P@L/5	ESMFold CAMEO	Layers
ESM-2 8M	8M	10.45	15.9	-	6
ESM-2 35M	35M	~ 9.8	25.7	-	12
ESM-2 150M	150M	~ 9.0	33.3	-	30
ESM-2 650M	650M	~ 8.2	44.4	0.865	33
ESM-2 3B	3B	~ 7.4	52.0	-	36
ESM-2 15B	15B	6.37	54.5	0.871 (TM 72.1)	48

Table 3. ESM-2 validation perplexity and downstream metrics. CAMEO = CAMEO structure benchmark mean TM-score or LDDT depending on source. Contact P@L/5 = long-range precision.

5.2 Compute-Optimal Protein Scaling

The NeurIPS 2024 study by Cheng et al. [14] derived compute-optimal scaling laws for protein LMs across 300+ models from 3.5M to 10.7B parameters: the optimal model size scales as $N_{\text{opt}} \sim C^{0.77}$ and optimal tokens as $D_{\text{opt}} \sim C^{0.23}$ (MLM objective). The contrast with Chinchilla's text exponents ($\sim 0.50/0.50$) is striking: protein MLMs should be far more parameter-heavy relative to data tokens than text LLMs. A corollary is that ESM-2 3B (trained on $\sim 1T$ tokens, ~ 45 epochs of UniRef50) is severely overtrained -- a compute-optimal allocation would train a 10.7B model on 260B unique tokens and match or exceed ESM-2 3B's performance at lower total FLOPs.

A competing analysis by Serrano et al. [15] trained 35M-parameter models on yearly UniRef100 snapshots (2011-2024) and found $N_{\text{opt}} \sim C^{0.27}$, $D_{\text{opt}} \sim C^{0.71}$ -- the opposite recommendation. The disagreement likely reflects dataset composition: Cheng et al. used UniMeta200B with 52.5% metagenomic sequences (high diversity), while Serrano et al. used UniRef-only data (increasing redundancy). This implies the compute-optimal allocation is

itself data-composition-dependent -- a dataset of diverse metagenomes prescribes larger models, while a dataset of redundant curated sequences prescribes more data.

5.3 ProteinGym Scaling Ceiling

Zero-shot fitness prediction on ProteinGym does not follow the perplexity scaling law beyond ~650M-4B parameters. ESM-2 3B and 15B do not improve over 650M on ProteinGym zero-shot Spearman. ESM-3 at 98B achieves 0.466 -- comparable to ESM-2 650M (0.414) and substantially below the current SOTA (AIDO Protein-RAG 16B: 0.518). Most strikingly, a parameter-free evolutionary statistics method, RSALOR (Riesselman-style MSA + structural contacts), achieves Spearman rho = 0.582 -- outperforming all neural models at any scale, including 40B-parameter Evo 2 (rho = 0.360) [16]. Of 46 downstream tasks, only 39% show predictable monotonic scaling behaviour; the remainder are nonmonotonic, inverse, or trendless.

6. Does Data Scale-Out Work?

6.1 UniRef Diversity Ratio Decline

UniRef100 has grown 33-fold from 11.7M (2011) to 390.8M (2024) sequences, yet performance gains from training on newer snapshots are non-monotonic. AMPLIFY's scaling study [11] finds a consistent performance drop between 2018 and 2021 UniRef100 snapshots despite adding ~130M sequences. The UniRef90/UniRef100 diversity ratio declines each year, indicating that new sequences are increasingly redundant with existing ones. Redundant training epochs waste compute and can degrade performance via effective overfitting. The implication is that naive data accumulation within the UniRef paradigm has saturated -- diversity, not volume, is the limiting factor.

6.2 Metagenomics Breaks the Saturation

Incorporating metagenomic sequences breaks UniRef saturation. ProtBERT-BFD (trained on 2.1B BFD sequences including metagenomics) outperforms ProtBERT (216M UniRef100) on all tasks. ESM C's JGI-augmented training (2B clusters, 32% of 6.2T tokens) enables 300M ESM C to match 650M ESM-2. The NeurIPS 2024 compute-optimal study used UniMeta200B (52.5% metagenomic), and a 2025 bioRxiv preprint claims global metagenomics adds ~4x the currently known protein family space. Yet MGnify alone has >5.7B non-redundant sequences, of which 1.17B have no similarity to any reference genome or Pfam family -- the 'dark proteome.' Only a fraction of this has been included in any published PLM training run.

6.3 Dataset Taxonomy

Dataset	Type	Size	Used by (models)	Key note
UniRef50	Protein seq	53-65M seqs, ~.	ESM-2, ESM-1b, ProtGPT2	ESM-2 uses as fram..
UniRef90	Protein seq	98M-250M seqs	ESM-1v, ProGen2, xTrimoPGLM	More diverse than ..
UniRef100	Protein seq	11.7M(2011)-39..	ProtBERT, RITA, Tranception,..	33x growth; divers..
BFD	Protein seq	2.1B seqs, 393..	ProtBERT-BFD, ProtT5-XL, Pro..	Metagenomic; BFD-t..
MGnify	Protein seq	>5.7B seqs (gr..	ESM C (372M clusters)	1.17B truly novel ..
JGI	Protein seq	2B clusters at..	ESM C only	Only PLM to use JG..
UniMeta200B	Protein seq	939M seqs, 194..	Compute-Optimal NeurIPS 2024	52.5% metagenomic;..
AFDB (40M)	Structure	~40M AF2 predi..	SaProt (3Di tokens)	3Di tok: structure..
ESMAtlas	Structure	617M predicted..	ESM-3	Metagenomic dark p..
hg38	DNA	3.2B nucleotides	DNABERT, HyenaDNA, Caduceus	Single genome; suf..
1000 Genomes	DNA	3,202 human ge..	NT v1 (1000g variants)	Captures populatio..

NT-multi (850)	DNA	174B nt, 850 g..	NT v1, NT v2	Multi-species cros..
DNABERT-2 multi	DNA	32.5B nt, 135 ..	DNABERT-2	BPE tokenization; ..
OpenGenome	DNA	300B tokens	Evo (Evo 1)	Prokaryote+phage o..
OpenGenome2	DNA	8.85T nucleoti..	Evo 2	All 3 domains of l..
ProteinGym	Annotation	217 assays, 2...	Benchmark (all major models)	De facto standard;..

Table 4. SI = sequence identity. AFDB = AlphaFold Database. UniMeta200B is the only purpose-built compute-optimal corpus; all others were constructed for prior research goals.

7. The Bottleneck: Where the Ceiling Is

Synthesising the evidence above, the field faces five compounding bottlenecks. None is primarily a compute or architecture problem.

- **DATA REDUNDANCY IN UNIREF:** The dominant protein training corpus (UniRef) adds increasingly similar sequences each year. Training on newer UniRef100 releases can degrade model performance due to effective overfitting on near-identical sequences. Metagenomics (BFD, MGnify, JGI) breaks this ceiling but has not been fully incorporated into any training corpus. The 1.17B MGnify dark proteome sequences -- genuinely novel, no Pfam hit -- represent the largest unexploited reservoir of protein diversity.
- **ANNOTATION GAP:** Unannotated protein sequences grow exponentially (metagenomics), while experimentally annotated sequences (SwissProt GO/EC terms, DMS fitness assays) grow only linearly. All supervised fine-tuning, fitness prediction, and pathogenicity classification tasks depend on this shrinking annotated fraction. No current model architecture solves this -- it is a data collection problem.
- **PROTEINGYM SATURATION AT ~4B PARAMETERS:** Zero-shot fitness prediction peaks at ~650M-4B parameters and declines with further scaling. A parameter-free MSA method (RSALOR, $\rho=0.582$) outperforms the 40B-parameter Evo 2 ($\rho=0.360$). This ceiling is consistent with the theoretical limit on pairwise epistasis encoding and implies that overcoming it requires rethinking the training objective, not scaling the existing one.
- **GAIN-OF-FUNCTION BLIND SPOT:** All conservation-based PLMs are systematically miscalibrated for GOF disease proteins. The LewyGym finding (ESM-2 $\rho=-0.17$ on SNCA vs $+0.414$ on LOF proteins) quantifies this for Parkinson's disease. No existing training dataset provides aggregation-specific DMS data at scale that would allow models to learn the GOF signal. This is simultaneously a data gap and a training objective gap.
- **RNA AND EUKARYOTIC DARK MATTER:** No foundation model analogous to ESM-2 exists for RNA (despite 27M+ non-redundant sequences in RNACentral and >4,000 Rfam families). Eukaryotic metagenomics (protists, micro-algae, fungi) is one order of magnitude less sequenced than prokaryotic metagenomics. Evo 2 is the first genomic LM to include eukaryotes at scale, with only months of published results.

8. Untapped Training Data

Table 5 summarises data sources that exist but have not been fully incorporated into any published PLM training run, ranked by estimated impact.

Source	Scale	Why valuable / why unused
MGnify dark proteome (no Pf..)	1.17B seqs	Truly novel folds/functions, globally diverse. Only fract..
Global metagenomics expansion	~4x known protei..	2025 preprint claims 4x protein family space in metagenom..
JGI (soil/freshwater/marine..)	2B clusters at 7..	Only ESM C uses JGI. Orthogonal diversity to MGnify (diff..)
Viral proteomes (eukaryote-..)	>330K viral geno..	Excluded by design from Evo 1/2. Viral proteins have dist..
SNCA/PD aggregation DMS (to..)	~2,660 SNCA vari..	Direct aggregation measurements would enable GOF-calibrat..

RNA sequences (RNACentral, ..	>27M non-redunda..	No RNA foundation model exists. Non-coding RNAs are >90% ..
Eukaryotic metagenomics (Me..	Tens of millions..	Eukaryotic microbes (protists, algae, fungi) are 10x unde..
Structural annotations via ..	>5B seqs remaining	SaProt shows 3Di tokens double contact prediction perform..

Table 5. *Evo 1/2 explicitly excludes eukaryote-infecting viruses 'by design.'* SNCA aggregation DMS does not yet exist publicly at the scale needed for model training.

9. Discussion and Outlook

The biological language model field has achieved remarkable progress in structural representation (ESMFold TM-score 0.72 on CAMEO) and genomic modelling (Evo 2 1M-bp context), but has encountered a systematic ceiling in the task most directly relevant to human disease: zero-shot variant effect prediction. The ceiling is mechanistically interpretable: PLMs trained on evolutionary conservation encode pairwise sequence statistics that correlate with fitness for loss-of-function proteins but invert for gain-of-function disease proteins. No amount of scaling within the current training paradigm resolves this.

The path forward bifurcates. For GOF proteins specifically (SNCA/PD, SOD1/ALS, APP/Alzheimer's), the required training signal is aggregation-specific DMS data -- a data generation problem, not a model architecture problem. For the broader fitness prediction ceiling, the evidence points toward rethinking training objectives to encode higher-order epistasis rather than pairwise correlations. SaProt demonstrates that vocabulary design (441 SA tokens vs 20 amino acids) is a tractable efficiency lever. RSALOR's dominance on ProteinGym suggests explicit structural contact information, not just sequence statistics, is the missing inductive bias.

The genomic track faces a different frontier: eukaryotic regulatory biology at chromosome scale. Evo 2 has established 1M-bp context and eukaryotic pretraining; the challenge is now accumulating sufficient annotated eukaryotic variant data (regulatory element function, splicing QTLs, enhancer-promoter links) to fine-tune these models for clinical variant interpretation.

References

- [1] Lin, Z. et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* 379, 1123-1130 (2023). doi:10.1126/science.ade2574
- [2] Meier, J. et al. Language models enable zero-shot prediction of the effects of mutations on protein function. *NeurIPS* 2021. doi:10.1101/2021.07.09.450648
- [3] Hayes, T. et al. Simulating 500 million years of evolution with a language model. *Science* 2024 (ESM-3). doi:10.1101/2024.07.01.600583
- [4] Notin, P. et al. ProteinGym: Large-Scale Benchmarks for Protein Fitness Prediction and Design. *NeurIPS* 2023. doi:10.1101/2023.12.07.570727
- [5] Unsal, S. et al. Learning functional properties of proteins with language models. *Nature Machine Intelligence* 2022.
- [6] Consens, M. et al. To transformers and beyond: large language models for the genome. *arXiv* 2023.
- [7] O'Leary, E. & O'Leary, N. LewyGym: A Parkinson's Disease Protein Variant Effect Benchmark. *bioRxiv* 2026.
- [8] Elnaggar, A. et al. ProtTrans: Toward Understanding the Language of Life Through Self-Supervised Learning. *IEEE TPAMI* 2022. doi:10.1109/TPAMI.2021.3095381
- [9] Hesslow, D. et al. RITA: a Study on Scaling Up Generative Protein Sequence Models. *arXiv:2205.05789* (2022).
- [10] Nijkamp, E. et al. ProGen2: Exploring the Boundaries of Protein Language Models. *Cell Systems* 2023. doi:10.1016/j.cels.2023.10.002
- [11] Fournier, Q. et al. Protein Language Models are Scalable and Compute-Efficient Sequence Encoders. *AMPLIFY*. *bioRxiv* 2024.
- [12] Nguyen, E. et al. HyenaDNA: Long-Range Genomic Sequence Modeling at Single Nucleotide Resolution. *NeurIPS* 2023.

- [13] Nguyen, E. et al. Sequence modeling and design from molecular to genome scale with Evo. Science 2024. doi:10.1126/science.ad09336
- [14] Cheng, X. et al. Compute-Optimal Training Strategies for Protein Language Models. NeurIPS 2024.
- [15] Serrano, E. et al. Scaling and Data Saturation in Protein Language Models. arXiv:2507.22210 (2025).
- [16] Laine, E. et al. RSALOR: Rapid Structure-Augmented Log-Odds Ratio for variant effect prediction. bioRxiv 2024.
- [17] Su, J. et al. SaProt: Protein Language Modeling with Structure-Aware Vocabulary. ICLR 2024.
- [18] Cheng, J. et al. AlphaMissense: Accurate proteome-wide missense variant effect prediction. Science 2023. doi:10.1126/science.adg7492
- [19] Fraternali, F. et al. EVE: Evolutionary model of Variant Effect. Nature 2021. doi:10.1038/s41586-021-04043-8